

TEAM APEX

RESQAI

**AI-Assisted Disaster Response Through
Humanitarian Message Classification**

A Machine Learning Based Approach for
Real-Time Crisis Information Analysis

PROJECT REPORT

Submitted in partial fulfillment of the requirements
for the academic project

TEAM MEMBERS

Dhwani Agarwal	1024031024
Sanchita Jain	1024031020
Yuvakshi Sood	1024031025

**DEPARTMENT OF COMPUTER SCIENCE
THAPAR INSTITUTE OF ENGINEERING AND
TECHNOLOGY**

Abstract

Natural disasters and humanitarian emergencies generate large amounts of information through social media and other online communication channels. During such events, affected individuals may report injuries, request assistance, provide information about infrastructure damage, communicate evacuation requirements, or offer rescue and volunteering support. However, these useful messages are mixed with irrelevant, political, emotional, and general informational content. Manually filtering this information is time-consuming and difficult during rapidly evolving emergencies.

This project presents **ResQAI**, an AI-assisted disaster information processing system designed to automatically classify crisis-related social-media messages into humanitarian categories. The system uses **BERTweet**, a transformer-based language model pretrained specifically for Twitter-style text, and fine-tunes it on the **HumAID** dataset.

Multiple approaches were investigated, including traditional TF-IDF-based machine learning models, Logistic Regression, Linear Support Vector Machines, Random Forest, DistilBERT, and BERTweet. Among the evaluated approaches, BERTweet produced the strongest overall performance. The final BERTweet model achieved **78.67% accuracy** and a **78.27% weighted F1 score** on the unseen test set.

Error analysis revealed that several humanitarian categories have substantial semantic overlap. In particular, the `other_relevant_information` category was frequently confused with categories such as `caution_and_advice` and `not_humanitarian`. To address the limitations of a single classification model, a hybrid architecture is proposed in which BERTweet acts as the primary fast classifier while a large language model such as Gemini can provide secondary contextual analysis for low-confidence messages.

The proposed system is intended as a decision-support and information filtering system rather than an autonomous emergency-response authority.

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1 Introduction

Natural disasters such as earthquakes, floods, hurricanes, wildfires, and other humanitarian emergencies can create rapidly changing situations in which timely information is extremely valuable. During such events, people frequently use social media to report incidents, request assistance, share warnings, communicate their condition, or provide information about affected areas.

Although social media can provide valuable real-time situational information, the sheer volume of messages makes manual processing impractical. A stream of disaster-related messages may contain emergency requests alongside political discussions, expressions of sympathy, general news, jokes, opinions, and unrelated content.

ResQAI addresses this problem by automatically classifying incoming messages into humanitarian information categories. The purpose of the system is to filter and organize large amounts of information so that potentially useful messages can be identified more efficiently.

1.1 Problem Statement

Given a social-media message associated with a disaster or humanitarian event, the system should determine the most relevant humanitarian category of the message.

The classification categories include:

- reports of injured or dead people,
- infrastructure and utility damage,
- requests for urgent assistance,
- displaced people and evacuation information,
- missing or found people,
- rescue, volunteering, and donation efforts,
- caution and safety information,
- sympathy and support,
- other relevant disaster information, and
- non-humanitarian information.

1.2 Objectives

The primary objectives of ResQAI are:

1. To develop an automated classifier for humanitarian disaster messages.
2. To investigate both classical machine learning and transformer-based approaches.
3. To fine-tune a social-media-specific transformer model for disaster message classification.
4. To evaluate model performance using validation and unseen test data.
5. To identify and analyze major sources of classification errors.
6. To design a confidence-based architecture capable of using an LLM for ambiguous messages.

1.3 Motivation

During a disaster, emergency responders may receive thousands of messages within a short period of time. Not every message requires immediate action, and important messages can be buried among large quantities of less useful content.

An automated classification layer can therefore act as an initial information filter. Rather than replacing emergency personnel, such a system can help organize incoming information and highlight potentially relevant categories.

2 Background and Related Technology

2.1 Natural Language Processing

Natural Language Processing (NLP) is a field of artificial intelligence concerned with enabling computers to process and analyze human language.

Social-media text presents several challenges compared with formal written language. These include:

- abbreviations and informal expressions,
- spelling variations,
- hashtags and mentions,
- emojis and emoticons,
- incomplete sentences,

- noisy punctuation, and
- context-dependent meanings.

These characteristics make social-media-specific language models particularly useful for this application.

2.2 TF-IDF

Traditional machine learning experiments in this project used Term Frequency–Inverse Document Frequency (TF-IDF) to convert text into numerical vectors.

The TF-IDF value of term t in document d can be expressed as:

$$TFIDF(t, d) = TF(t, d) \times IDF(t) \quad (1)$$

where:

$$IDF(t) = \log \left(\frac{N}{df(t)} \right) \quad (2)$$

Here, N is the total number of documents and $df(t)$ is the number of documents containing term t .

TF-IDF provides an efficient baseline representation but does not directly model contextual relationships between words.

2.3 Transformer Models

Transformer architectures use self-attention mechanisms to capture contextual relationships between tokens. Transformer-based language models can therefore produce contextual representations that are more suitable for understanding natural language than simple frequency-based representations.

2.4 BERTweet

BERTweet is a transformer-based language model pretrained specifically on English Twitter data. Since ResQAI targets short and informal social-media messages, BERTweet is particularly suitable for the application.

The pretrained BERTweet model was fine-tuned using the HumAID dataset for ten-class humanitarian classification.

3 Dataset

The project uses the **HumAID** dataset, a humanitarian information dataset containing disaster-related social-media messages.

3.1 Dataset Distribution

The dataset used in the experiments consisted of:

Table 1: HumAID Dataset Distribution

Dataset Split	Number of Messages
Training	53,531
Validation	7,793
Testing	15,160
Total	76,484

3.2 Classification Categories

The dataset contains ten humanitarian information categories.

Table 2: HumAID Classification Categories

Class	Description
caution_and_advice	Safety warnings and recommendations
displaced_people_and_evacuations	Information about displaced people and evacuation
infrastructure_and_utility_damage	Reports of damage to buildings, roads, utilities, and other infrastructure
injured_or_dead_people	Reports concerning injured or deceased individuals
missing_or_found_people	Reports of missing or found individuals
not_humanitarian	Disaster-related content that is not directly humanitarian
other_relevant_information	Other relevant information concerning the disaster

Class	Description
<code>requests_or_urgent_needs</code>	Requests for emergency assistance, resources, or supplies
<code>rescue_volunteering_or_donation_effort</code>	Rescue activities, volunteer efforts, and donation-related information
<code>sympathy_and_support</code>	Messages expressing sympathy, solidarity, or emotional support

4 Methodology

The project followed an iterative machine learning development process.

4.1 Data Preparation

The dataset was loaded and divided into training, validation, and test sets. The class names were converted into numerical identifiers for model training.

Each class was assigned a unique integer:

$$label \rightarrow \{0, 1, \dots, 9\} \quad (3)$$

The text was then processed using the BERTweet tokenizer.

4.2 Tokenization

BERTweet tokenization converts each message into a sequence of token IDs and an associated attention mask.

The maximum sequence length was set to 128 tokens.

Before selecting this value, token-length statistics were examined. The average training and validation message lengths were approximately 35 tokens. Only approximately 0.10% of training messages and 0.04% of validation messages reached the 128-token limit.

Therefore, increasing the maximum sequence length was not expected to provide a meaningful improvement and would have increased computational requirements.

4.3 Model Training

The final BERTweet model was fine-tuned using the following configuration.

Table 3: Final BERTweet Training Configuration

Parameter	Value
Base model	BERTweet-base
Maximum sequence length	128
Learning rate	1×10^{-5}
Training batch size	8
Evaluation batch size	8
Maximum epochs	3
Weight decay	0.01
Warm-up steps	500
Learning-rate scheduler	Linear
Mixed precision	FP16
Model selection metric	Weighted F1
Early stopping patience	1

4.4 Model Selection

The model was evaluated at the end of every epoch. The checkpoint with the highest validation weighted F1 score was retained.

This approach is important because lower training loss does not necessarily mean better generalization. During experimentation, validation performance stopped improving after approximately the second epoch while training loss continued to decrease.

The use of early stopping and best-checkpoint selection therefore reduced the risk of retaining an overfit model.

5 Model Experiments

Several approaches were evaluated before selecting BERTweet.

5.1 Traditional Machine Learning Models

TF-IDF representations were used with classical classifiers including:

- Naive Bayes,
- Logistic Regression,
- class-weighted Logistic Regression,
- Linear Support Vector Machine, and

- Random Forest.

These models provided useful baselines but generally performed below the transformer-based models.

5.2 Transformer Models

Two transformer architectures were investigated:

- DistilBERT,
- BERTweet.

Class weighting was also investigated for DistilBERT and BERTweet-related experiments. However, class weighting did not improve the final performance and was therefore not used in the selected BERTweet configuration.

5.3 Model Comparison

The approximate experimental results are summarized below.

Table 4: Comparison of Evaluated Approaches

Model	Approx. Accuracy	Observation
Naive Bayes	~68–69%	Baseline
Logistic Regression	~73–75%	Strong classical baseline
Weighted Logistic Regression	~73%	No major improvement
Linear SVM	~72–73%	Competitive classical model
Random Forest	~71–72%	Lower performance
DistilBERT	~77%	Strong transformer baseline
Weighted DistilBERT	~76%	Weighting reduced performance
BERTweet	~79%	Best overall approach

The results indicate that transformer-based models were better suited to capturing the contextual characteristics of disaster-related social-media messages.

6 Results

6.1 Validation Performance

The selected BERTweet configuration showed the following validation behavior:

Table 5: BERTweet Validation Performance by Epoch

Epoch	Validation Accuracy	Weighted F1
1	76.90%	75.64%
2	78.40%	78.05%
3	78.28%	77.78%

The second epoch achieved the highest validation weighted F1 score.

The third epoch showed a small decline in validation F1 despite continued improvement in training loss. This indicates the beginning of overfitting and supports the use of best-checkpoint selection.

6.2 Final Test Performance

After model selection, the model was evaluated on the unseen test set.

Table 6: Final BERTweet Test Results

Metric	Result
Test Accuracy	78.67%
Weighted F1 Score	78.27%

The relatively close validation and test performance indicates that the selected model generalizes reasonably well to unseen data.

7 Error Analysis

A confusion matrix was used to examine the behavior of the final BERTweet classifier.

The most significant source of confusion was the `other_relevant_information` category. Messages belonging to this class were frequently predicted as other disaster-related categories.

This is understandable because the class itself acts as a broad category for relevant information that does not clearly belong to another humanitarian class.

7.1 Semantic Overlap

Examples of ambiguous messages include:

“Aftershocks pls #eqnz”

and:

“When will it be safe to drink water, have kids walk around barefoot?”

These messages contain useful disaster-related information, but their exact category can depend on context.

For example, an aftershock-related message may be interpreted as:

- other relevant information,
- caution and advice, or
- an urgent situation.

Similarly, a question about drinking water may represent a safety warning, an information request, or general disaster information.

7.2 Main Sources of Error

The primary sources of confusion observed during analysis were:

1. `other_relevant_information` versus `not_humanitarian`,
2. `other_relevant_information` versus `caution_and_advice`,
3. `requests_or_urgent_needs` versus `rescue_volunteering_or_donation_effort`.

These errors indicate that the performance limitation is not necessarily caused by insufficient model capacity. The category definitions themselves have overlapping semantic boundaries.

8 Proposed ResQAI Architecture

The final ResQAI concept extends the trained BERTweet classifier with a confidence-based secondary analysis layer.

8.1 Primary Classifier

BERTweet acts as the primary classification model.

For an incoming message, the model produces:

- predicted humanitarian category,
- prediction confidence.

High-confidence predictions can be processed directly by the system.

8.2 LLM Fallback

Messages for which BERTweet produces low-confidence predictions can be sent to a large language model such as Gemini for additional contextual analysis.

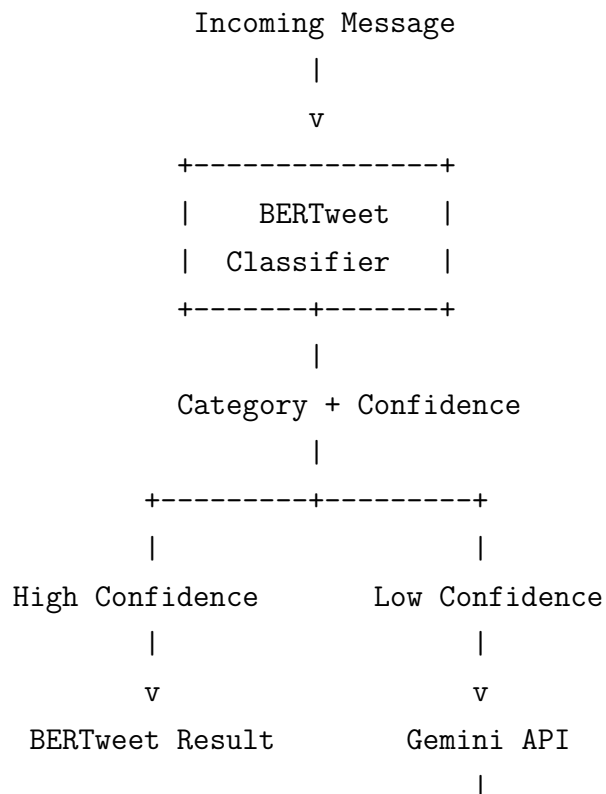
The LLM can be instructed to return structured information such as:

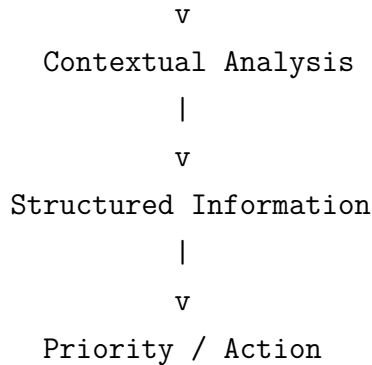
- humanitarian category,
- priority,
- location,
- number of affected people,
- type of assistance required,
- recommended action,
- explanation.

This component is proposed as a secondary reasoning layer rather than as a replacement for the trained classifier.

8.3 Hybrid Architecture

The proposed architecture is:





The architecture allows the domain-specific classifier to handle clear messages efficiently while providing additional reasoning for ambiguous messages.

9 System Workflow

The proposed workflow is as follows:

1. Receive an incoming disaster-related social-media message.
2. Tokenize the message using the BERTweet tokenizer.
3. Run the trained BERTweet classifier.
4. Obtain the predicted humanitarian category and confidence score.
5. Compare the confidence score with a predefined threshold.
6. If the confidence is sufficiently high, use the BERTweet prediction.
7. If the confidence is low, send the message to the secondary LLM analysis layer.
8. Extract structured information such as location, urgency, and assistance requirements.
9. Present the resulting information to the user or emergency information interface.

10 Example System Operation

Consider the following incoming message:

“People are trapped inside the collapsed building and urgently need rescue.”

The BERTweet classifier may identify the message as:

Category: injured_or_dead_people

Confidence: High

Since the message is classified with high confidence, the system can use the BERTweet prediction directly.

Now consider a more ambiguous message:

“Not sure if I slept through the aftershocks or if they are starting again.”

If the classifier produces a low-confidence prediction, the system can invoke the secondary LLM layer. The LLM can provide additional contextual interpretation and structured information such as the likely category, location, urgency, and recommended action.

This demonstrates why the proposed architecture does not simply send every message directly to an LLM.

11 Advantages

The proposed ResQAI architecture provides several advantages.

11.1 Domain-Specific Classification

BERTweet is fine-tuned specifically on humanitarian disaster messages rather than being used as a generic text classifier.

11.2 Social-Media Suitability

BERTweet was pretrained on Twitter-style data, making it appropriate for informal and short messages.

11.3 Reduced LLM Dependence

High-confidence messages can be processed directly by the trained classifier, reducing unnecessary external LLM calls.

11.4 Confidence-Based Reasoning

The system can reserve deeper LLM analysis for ambiguous messages.

11.5 Scalability

Automated classification can process large volumes of messages more efficiently than manual inspection.

11.6 Decision Support

The system is designed to organize information and support human decision making rather than independently making emergency decisions.

12 Limitations

The current implementation has several limitations.

1. The final classifier achieves approximately 79% accuracy rather than perfect classification.
2. Several humanitarian categories have overlapping semantic meanings.
3. Social-media language is highly informal and context-dependent.
4. The model is primarily trained on English-language data.
5. A confidence threshold cannot guarantee that all ambiguous messages will be correctly interpreted.
6. The proposed LLM fallback architecture requires external API access.
7. Real-world emergency deployment would require extensive reliability, security, privacy, and human-oversight testing.

Therefore, ResQAI should be treated as an information filtering and decision-support system rather than an autonomous emergency-response system.

13 Future Scope

Several improvements can be explored in future versions of ResQAI.

13.1 Multilingual Support

Disaster communication may occur in multiple languages. Multilingual transformer models could be introduced to support regional and international disaster communication.

13.2 Location Extraction

A dedicated entity extraction component could identify locations mentioned in messages and map them to geographical coordinates.

13.3 Real-Time Message Streams

The classifier could be integrated with real-time social-media or emergency message streams to support continuous monitoring.

13.4 Priority Prediction

A separate model could predict the urgency of a message independently from the humanitarian category.

13.5 Multimodal Analysis

Images and videos shared during disasters could be analyzed together with textual information.

13.6 Human-in-the-Loop Verification

Messages identified as critical could be presented to trained emergency operators for verification before any operational action is taken.

13.7 Improved Classification Boundaries

Further work could investigate hierarchical classification or multilabel classification because some disaster messages naturally contain information belonging to multiple categories.

14 Conclusion

This project presented **ResQAI**, an AI-assisted system for processing and classifying humanitarian disaster messages.

Multiple machine learning and transformer-based approaches were evaluated. Traditional TF-IDF-based approaches provided useful baselines, while transformer-based approaches produced stronger performance. BERTweet was selected as the primary model because of its suitability for social-media language and its superior performance in the experiments.

The final BERTweet model achieved:

78.67% Test Accuracy

78.27% Weighted F1 Score

on the unseen test set containing 15,160 messages.

Error analysis showed that the largest challenge was semantic overlap between humanitarian categories, particularly the broad `other_relevant_information` category and related classes.

The project therefore proposes a hybrid architecture in which BERTweet acts as the primary domain-specific classifier and a large language model can be used as a secondary reasoning layer for low-confidence messages. This architecture combines the efficiency and measurable performance of a specialized classifier with the contextual reasoning capabilities of an LLM.

ResQAI is ultimately intended to help organize the large volume of information generated during disasters and surface potentially important information for further human assessment.

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